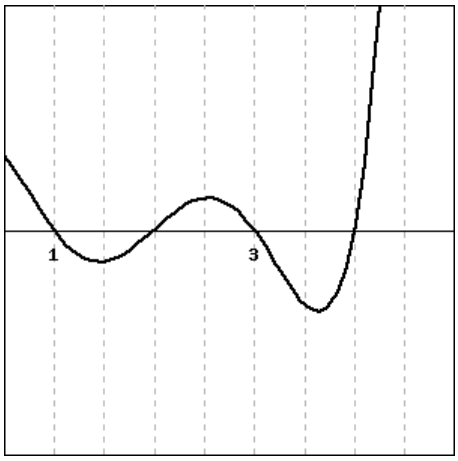
Worksheet 5 - The Derivative

MATH 2263

09/22/2026

1. Consider the function $y = f(x)$ graphed below:



Using this graph (and estimating your answer whenever necessary), give the x-coordinate of a point where...

(a) the **derivative** of $f(x)$ is negative:

(b) the **value** of $f(x)$ is negative:

(c) the **derivative** of $f(x)$ is smallest (most negative):

(d) the **derivative** of $f(x)$ is zero:

(e) the **derivative** of $f(x)$ is approximately the same as the **derivative** of $f(x)$ at $x = 2.25$:

2. Suppose that a ball is tossed straight up into the air. At time t (the number of seconds after the ball was thrown), the height of the ball (in feet) is given by the function

$$s(t) = 64 - 16(t-1)^2.$$

(a) Sketch the graph of $s(t)$ on the interval $[0, 3]$. Label the points on the graph that correspond to when the ball was released, when the ball reaches its highest point, and when the ball lands.

(b) Compute the value of $\frac{s(2.5)-s(1.5)}{2.5-1.5}$.

(c) What are the units on your answer to part (b)? _____

(d) Within the context of this problem (i.e., the motion of the ball), what is the meaning of the value that you found in (b)?

(e) The value that you found in (b) also represents the slope of a certain line. Draw this line on the graph you drew in part (a).

(f) Use a calculator to complete the following table:

Time interval $[a, b]$	$\frac{s(b)-s(a)}{b-a}$
$[1.9, 2]$	
$[1.99, 2]$	
$[2, 2.1]$	
$[2, 2.01]$	

(g) Using your completed table, estimate the *instantaneous* velocity of the ball at time $t = 2$.

(h) Compute the derivative of the function $s(t)$ at $t = 2$. Compare this to your estimate above; is what you notice surprising?

3. Compute the derivatives of the following functions at the indicated points:

(a) $f(x) = x^2$ at $x = -1$.

(b) $f(x) = \dfrac{1}{x}$ at $x = 2$.

(c) $f'(1)$ where $f(x) = \sqrt{x+3}$.

(d) $f'(1)$ where $f(x) = \dfrac{1}{x^2+1}$.

(e) $f'(1)$ where $f(x) = x^3 + x - 1$.

(f) $f'(0)$ where $f(x) = \sin(x)$.